Talha Tallat
11-9-2026

Brief Requirements

JUTT CARS LTD is a car importation and sales business specialising in importing quality used passenger vehicles from Japan for resale in Ireland. The business primarily targets German and Japanese manufacturers, including Volkswagen, Audi, BMW, Toyota, Nissan and Honda, to meet the demand of the Irish market.

Vehicles are purchased from Japanese auctions, exporters and third-party suppliers before being shipped to Ireland. Although the business may expand to include 40-foot container shipping, which can accommodate up to six vehicles depending on their size and loading arrangements, the initial database implementation will focus exclusively on Roll-on/Roll-off (RoRo) shipping. The company also accommodates customer requests and pre-orders for specific vehicles.

Upon arrival in Ireland, imported vehicles undergo customs clearance, which involves an AIS agent submitting the necessary customs declarations and facilitating the release process. Following customs clearance and shipment release, a transport contractor collects the vehicles from the port and delivers them to the company's designated premises.

Once the vehicles have arrived at the company's premises, they undergo the applicable Vehicle Registration Tax (VRT) inspection and Irish registration procedures before being offered for sale. Following registration, vehicles undergo an NCT inspection to assess their roadworthiness. If an NCT inspection identifies defects requiring attention, the vehicle may be sent to a garage for the necessary repairs and subsequent inspection where required.

The company engages external VRT contractors to transport vehicles to the relevant registration facilities and facilitate the inspection process. Maintaining vehicle registration, inspection, repair and contractor records is important for operational tracking and year-end accounting requirements.

The company requires a relational database management system to maintain accurate records of its suppliers, vehicle purchases, shipping consignments, imported stock, vehicle specifications, Irish vehicle registration, contractors, customs clearance, NCT inspections, repairs, garages, customers and vehicle sales.

The proposed system will provide a centralised approach to managing the vehicle importation and sales process, from purchasing a vehicle in Japan through shipping, customs clearance, registration and inspection to its eventual sale in Ireland.

Scope

The initial database implementation will focus on the following business processes:

1. **Supplier** Management: Maintaining overseas supplier information, including business names, supplier types, contact details and addresses.

2. Importation and Shipping: Recording vehicle **consignments**, shipping arrangements, ports of loading and discharge, and arrival dates.

3. **Stock** Management: Maintaining an inventory of imported vehicles and recording procurement information, auction details, purchase prices and vehicle condition.

4. **Vehicle** Management: Recording the specifications of imported vehicles, including manufacturer, model, year, chassis number, mileage, fuel type, transmission and colour.

5. **Shipment Release** and Customs Clearance: Recording shipment release information and customs clearance details when imported vehicles arrive and are offloaded at the Irish port of entry.

6. AIS **Agent** Management: Maintaining details of Authorised Intermediary Service (AIS) agents responsible for processing customs declarations and facilitating customs clearance for imported consignments.

7. **Transport Contractor** Management: Recording details of transport contractors engaged to collect and deliver vehicles from the port of arrival to the company's designated premises.

8. Vehicle **Registration**: Maintaining Irish vehicle registration records, including registration dates, registration numbers, registration locations and vehicle emissions information (NOx and CO_2).

9. VRT Contractor Management: Recording details of contractors engaged to transport vehicles to the Vehicle Registration Tax (VRT) inspection and **registration** facilities, including company and employee information.

10. NCT Management: Recording National Car Test (**NCT**) bookings and inspection-related information for individual vehicles.

11. Vehicle **Repairs** and Maintenance: Recording vehicle damage, repairs carried out and replacement parts where applicable.

12. **Garage** Management: Maintaining details of garages responsible for carrying out vehicle repairs and maintenance.

13. Vehicle Sales Management: Recording vehicle sales transactions, including the **customer**, vehicle sold, sale date.

Future development may include support for 40ft container shipping arrangements, financial transaction management, accounting integration and vehicle listings on online marketplaces.

Business Rules – Reactions

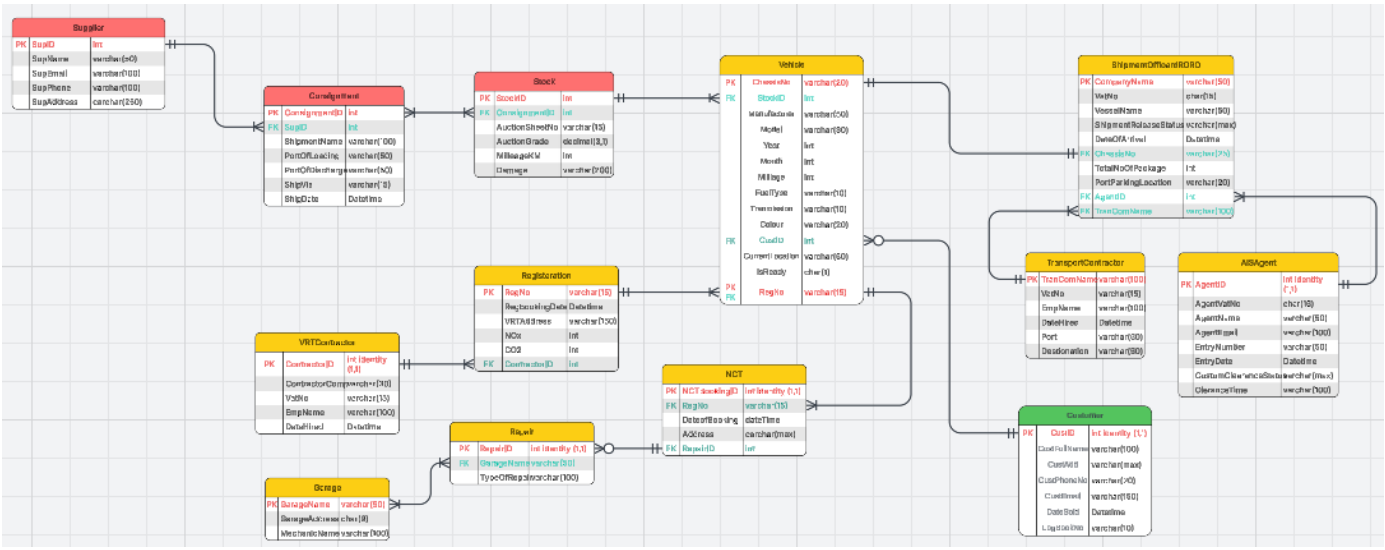
The business rules define how JUTT CARS LTD operates and how information must be maintained within the database. They also establish the integrity constraints and actions that apply when records are inserted, updated or deleted.

1. Supplier Management
    - Each supplier must have a unique Supplier ID.
    - A supplier may supply multiple vehicles over time.
    - Each consignment must be associated with exactly one supplier.
    - Suppliers contact details, including business name, email address, telephone number and address, should be maintained where available.
2. Importation and Consignment Management
    - Each consignment must have a unique Consignment ID.
    - Each consignment must reference an existing supplier.
    - Under the current RoRo shipping arrangement, each consignment contains one vehicle.
    - A supplier may be associated with multiple consignments.
    - Shipment information, including the vessel, ports of loading and discharge, shipping date and arrival date, should be recorded where available.
    - A consignment must not be deleted while an associated stock record exists.
3. Stock and Procurement Management
    - Each stock record must have a unique Stock ID.
    - Each stock record must reference one existing consignment.
    - Every vehicle purchased must have a corresponding stock record.
    - Each purchased vehicle must have an auction sheet or equivalent procurement reference recorded.
    - Vehicles must achieve a minimum auction grade of 4.0 to meet the company's purchasing criteria.
    - Vehicles must have a recorded mileage of less than 100,000 kilometres at the time of purchase.
    - Mileage cannot be negative, and auction grades must be within the valid grading range.
4. Vehicles
    - Each vehicle must have a unique chassis number.
    - Each vehicle must be associated with one existing stock record.
    - A stock record must not be associated with more than one vehicle.
    - Vehicle details, including manufacturer, model, year, mileage, fuel type, transmission and colour, etc should be maintained.

Implementation and Evaluation

- One supplier can supply multiple vehicles (for example, SBT Co. Ltd supplying 5 cars), therefore must have 5 consignments. Each consignment belongs to exactly one supplier. Each RoRo (Roll-On_Roll_Off) consignment contains one vehicle. Each stock record belongs to one consignment.
- Vehicle cannot exist without the stockID, therefore, must have stock ID. when the stock does not exist it must have no vehicle associated with it and cannot remove the stock when vehicle exist already.

ERD


Supplier
SupID
SupName
SupEmail
SupPhone
SupAddress
Consignment
ConsignmentID
SupID
ShipmentName
PortOfLoading
PortOfUnloading
ShipVia
ShipDate
Stock
StockID
ConsignmentID
AuctionStreetNo
AuctionGrade
MileageKM
Damage
Vehicle
ChassisNo
StockID
Make
Model
Year
Month
Mileage
FuelType
Transmission
Colour
CustID
CurrentLocation
IsReady
RegNo
ShipmentOffloadRORO
CompanyName
VatNo
VesselName
ShipmentReleaseStatus
DateOfArrival
ChassisNo
TotalNoOfPackage
PortParkingLocation
AgentID
TranComName
TranportContractor
TranComName
VatNo
EmpName
DateHired
Position
AISAgent
AgentID
AgentVatNo
AgentName
AgentEmail
EntryNumber
EntryDate
CustomClearanceStatus
ClearanceTime
Registeration
RegNo
RegistrationDate
VRTAddress
NCT
CO2
ContractorID
VRTContractor
ContractorID
ContractorCompany
VatNo
EmpName
DateHired
NCT
NCTBookingID
RegNo
DateofBooking
Address
RepairID
Repair
RepairID
GarageName
TypeOfRepair
Garage
GarageName
GarageAddress
MechanicName
Customer
CustID
CustFullName
CustVat
CustPhoneNo
CustEmail
DateSold
LogOnDate